Simon Rosen

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Professional Summary

Linguist specializing in phonetics, phonology, and prosody, working on speech and voice AI evaluation at ServiceNow since 2023. I design TTS and voice agent evaluations, including multilingual vendor comparisons, write the rubrics and annotation guidelines behind them, and build the internal labeling tools our linguists use to run them. Co-first author of *Beyond Naturalness* (EMNLP 2026 Main), a dimension-level TTS evaluation benchmark.

Core Competencies

Speech & Linguistics: Phonetics & Phonology, Prosody & Tone, Segmental/Suprasegmental Analysis, IPA Transcription, Acoustic Analysis (Praat), Forced Alignment (Montreal Forced Aligner), Pragmatics & Discourse

Voice AI Evaluation: TTS & ASR Evaluation, TTS Vendor Comparison, Text Normalization Test Design, MOS Predictor & Audio-LLM Judge Benchmarking, Voice Agent Conversation Rating, Failure Taxonomies, LLM-as-Judge Design

Experiment Design & Statistics: Benchmark Construction, Rubric & Annotation Guideline Design, Inter-Annotator Agreement (Krippendorff's α, Bootstrap CIs), Significance Testing (Kendall's τ, Mann–Whitney U, McNemar)

Engineering: Python (pandas, scikit-learn, statsmodels, matplotlib/seaborn), Internal Web Tool Development, LLM Applications & Prompt Engineering, Structured Output Schemas, AI-Assisted Development (Claude Code), JSON/YAML, Git

Professional Experience

Senior ML Data Linguist Feb 2026 – Present

ServiceNow, LUNAR (Linguistic Unit for NLU Analysis & Research) Team, Santa Clara, CA

- Co-designed a 10-dimension, linguistically grounded annotation schema that breaks TTS "naturalness" into word-level, prosodic, and paralinguistic dimensions, and built an 860-utterance benchmark labeled by trained linguist raters (EMNLP 2026 Main, co-first author).
- Compared eight automated speech evaluators (four MOS predictors, four audio-LLM judges) against human labels under four prompting conditions, showing per dimension where each evaluator type can and cannot replace human listeners.
- Designed and ran a multilingual TTS vendor comparison (Cartesia vs. ElevenLabs): wrote test sentences drawn from ServiceNow case language and normalization edge cases (symbols, math, units, URLs), defined a rubric pairing holistic 1–5 ratings with per-category head-to-head judgments, and adapted it with a native-speaker linguist for Japanese.
- Built a rating framework for AI voice agent conversations (speech fluency, tool-calling, task completion, helpfulness) and wrote its annotation guidelines.
- Ran the labeling efforts that produced the human benchmarks for customer-facing voice fluency judges: wrote the guidelines, onboarded labelers, and presented results across agents, separating ASR-driven failures (such as repeated confirmations) from conversation-flow and voice-style problems.
- Currently leading DocQnA success labeling and building the matching LLM judge.
- Built and maintain the internal labeling tools used on these projects, plus the team's project-tracking app.

ML Data Linguist Jan 2023 – Feb 2026

ServiceNow, LUNAR Team, Santa Clara, CA

- Wrote LLM-as-judge prompts and structured output schemas for internal evaluation tooling; authored a prompt design guide adopted as the team reference and led training sessions so product teams could build their own evaluations.
- Cataloged recurring failure patterns (hallucination, under-specification, intent misclassification) and worked with ML engineers on prompt changes, retraining, and production model selection.
- Built a ground-truth benchmark through paired human–model evaluation and checked it against LLM-as-judge and automatic metrics to find scoring discrepancies.

Publications

O. Bamgbose[*], **S. Rosen**[*], et al. *Beyond Naturalness: Probing Automated Text-to-Speech Evaluators on Linguistically Grounded Dimensions.* EMNLP 2026 (Main Conference). arXiv:2608.09930

[*]Equal contribution. Designed the annotation schema and led the annotation effort; dataset, schema, and evaluation code are publicly released.

Education

B.A. in Linguistics 2022

Princeton University, Princeton, NJ

Thesis: *Tone Sandhi and Constituency in Classifier Phrases Across Sinitic Languages*

Relevant Coursework: Phonetics & Phonology, Syntax, Semantics, Pragmatics, Java Programming, Data Structures & Algorithms